Cycles of length $2^n - 2$ in the binary de Bruijn graph

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Abstract

The binary de Bruijn graph $B(n)$ has $2^{2^{n-1}-n}$ cycles through all of its 2^n vertices. A count of the cycles that miss exactly two vertices, attributed to Kibler in 1972, appears in a 1982 survey with only an outline of a proof, and the formula printed there gives 4 for $n = 3$, where there are 3. We prove that the number is $\frac{1}{3}\big(2^n - (-1)^n\big)\, 2^{2^{n-1}-2n+2}$. The two missing vertices are either 0^n and 1^n or the two alternating strings. The first case is the de Bruijn count; in the second, the Laplacian changes by a matrix of rank one, and the determinant can be evaluated exactly because a random walk on the de Bruijn graph is uniform after $n-1$ steps. Evaluated at $n-1$, the printed expression is the count of the second case. The algebra is checked in the Lean proof assistant.

1 The question

Write 0s and 1s around a circle so that every block of three consecutive symbols is different. With eight symbols every one of the eight blocks of length three appears once; there are two such circles, 00010111 and 00011101, read cyclically. Now allow six symbols. How many circles are there? Stop and try this before reading on.

There are three: 000111, 001011 and 001101. The first misses the blocks 010 and 101; the other two are the circles of length eight with one 0 and one 1 removed from their runs of three, so they miss 000 and 111.

In graph terms, the vertices of the de Bruijn graph $B(n)$ are the binary strings of length n, with an edge from $a_1a_2\cdots a_n$ to $a_2\cdots a_nb$ for $b \in \{0,1\}$. A cycle of length L in $B(n)$ is the same as a circle of L symbols in which every block of n consecutive symbols occurs at most once. For $L = 2^n$ these are the de Bruijn cycles, and Flye Sainte-Marie [4] and de Bruijn [2] found that there are $2^{2^{n-1}-n}$ of them (de Bruijn's note [3] records the priority).

In 2016 Timothy Chow asked on MathOverflow [7] about a count of the cycles of length $2^n - 2$ that Fredricksen's survey of shift-register cycles [5, Section 2c] attributes to Kibler (personal communication, 1972). The survey gives only an outline: the BEST theorem reduces the count to a determinant, which "by induction" can be evaluated, and the total is printed as $\frac{1}{3}\,2^{2^n-2n}\big(2^{n-1} + (-1)^n\big)$ [5, p. 203], which is 4 for $n = 3$. Chow's question quotes it correctly; the slip is in the survey.

Theorem 1. *For $n \geq 2$ the de Bruijn graph $B(n)$ has*

$$T(n) = \frac{2^n - (-1)^n}{3}\, 2^{\,2^{n-1}-2n+2}$$

cycles of length $2^n - 2$. Of these, $2^{2^{n-1}-n}$ miss the vertices 0^n and 1^n, and

$$\tfrac{1}{3}\big(2^{n-2} - (-1)^n\big)\, 2^{2^{n-1}-2n+2}$$

miss the two alternating strings $0101\cdots$ and $1010\cdots$.

The first factor of $T(n)$ is the Jacobsthal number $1, 3, 5, 11, 21, \ldots$ and the second a power of two; the values for $n = 2, 3, 4, 5, 6$ are $1, 3, 20, 2816, 88080384$. The count of the second kind is the printed expression evaluated at $n - 1$.

2 Two edges left out

Put $m = n - 1$. A string of length n is an edge of $B(m)$, from its first m symbols to its last m, so the vertices of $B(n)$ are the edges of $B(m)$ and a cycle in $B(n)$ is a closed walk in $B(m)$ that uses no edge twice. A cycle of length $2^n - 2$ uses every edge of $B(m)$ except two; call the two unused edges R. The cycle is then an Eulerian circuit of $B(m) - R$.

Every vertex of $B(m)$ has two edges in and two out, and an Eulerian graph has as many edges in as out at every vertex, so the two edges of R also enter and leave each vertex equally often. Two edges with that property form either two loops or a cycle of length two.

Lemma 1. *The loops of $B(m)$ are at 0^m and 1^m, and its only cycle of length two joins the alternating strings $p = \cdots 0101$ and $q = \cdots 1010$.*

Proof. A loop at u means that shifting u by one place gives u back, so all its symbols are equal. If $u \to v \to u$ with $u \neq v$, shifting u by two places gives u back, so u has period two and is not constant. □

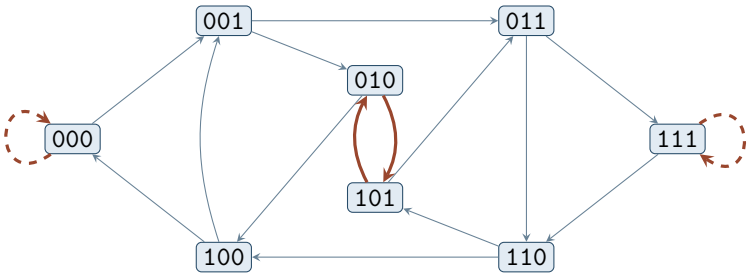


Figure 1: The de Bruijn graph $B(3)$. A cycle of length 14 in $B(4)$ is an Eulerian circuit of $B(3)$ with two edges removed: the two dashed loops (case A) or the two thick edges between 010 and 101 (case B).

By the BEST theorem [1, 8], the number of Eulerian circuits of a connected graph with in-degree equal to out-degree everywhere is the number of spanning arborescences rooted at any vertex w, times $\prod_v (d^+(v) - 1)!$. Here every out-degree is 1 or 2, so the product is 1 and each case counts arborescences. (If $B(m) - R$ is disconnected it has no arborescence and no Eulerian circuit, so the count is still right; this happens only in case B with $m = 1$.)

Case A. A loop is never part of an arborescence, so deleting the loops changes nothing and case A gives the de Bruijn number $2^{2^m - m - 1} = 2^{2^{n-1} - n}$. Directly: delete the two loop steps from a de Bruijn cycle.

3 The alternating case

Here is the route. Tutte's matrix-tree theorem [9] turns the number of arborescences into a determinant of the Laplacian. Deleting the two edges of the cycle $p \leftrightarrow q$ changes the Laplacian by a matrix of rank one, so by the matrix determinant lemma the ratio of the two tree counts is $1 - w^{\mathsf{T}} G w$ for one vector w and a matrix G that inverts the Laplacian. That inverse is a finite sum, because a random walk on $B(m)$ forgets where it started after exactly m steps.

Let $N = 2^m$, let P be the transition matrix of the walk that shifts in a uniformly random bit, and let J be the all-ones matrix. The Laplacian of $B(m)$ is $L = 2(I - P)$ (loops cancel).

Deleting $p \to q$ and $q \to p$ lowers the out-degrees of p and q by one and removes the two entries, so the new Laplacian is

$$L' = L - e_p(e_p - e_q)^{\mathsf{T}} - e_q(e_q - e_p)^{\mathsf{T}} = L - ww^{\mathsf{T}}, \qquad w = e_p - e_q.$$

Lemma 2. *If a matrix M has zero row sums and zero column sums, all its cofactors are equal to one number $c(M)$, and $\det(M + J) = N^2 c(M)$.*

Proof. If M has rank $N-2$ or less, every cofactor and $\det(M+J)$ vanish. Otherwise its kernel and left kernel are both spanned by $\mathbf{1}$, and $M \operatorname{adj}(M) = \operatorname{adj}(M)M = 0$ forces $\operatorname{adj}(M) = cJ$. The matrix determinant lemma in the form $\det(M + uv^{\mathsf{T}}) = \det M + v^{\mathsf{T}} \operatorname{adj}(M) u$, with $u = v = \mathbf{1}$, gives $\det(M + J) = 0 + c\,\mathbf{1}^{\mathsf{T}} J \mathbf{1} = N^2 c$. □

By the matrix-tree theorem the number of arborescences of each graph is the cofactor of its Laplacian, so the ratio we want is $\det(L' + J)/\det(L + J)$.

Lemma 3 (exact mixing)**.** *$P^m = J/N$. For $0 \le k < m$, $P^k(u, v) = 2^{-k}$ if v is u shifted by k places (with any k new symbols), and 0 otherwise.*

Proof. After k steps the last k symbols are uniform random and the rest are u shifted; after m steps nothing of u is left. □

Put $Z = \sum_{k=0}^{m-1} (P^k - J/N)$. Since $PJ = J$, the sum telescopes: $(I - P)Z = I - P^m = I - J/N$. Also $Z\mathbf{1} = 0$ and $\mathbf{1}^{\mathsf{T}} Z = 0$, because P has row and column sums 1. For $y = \frac{1}{2} Z w$ this gives $(L + J)y = (I - J/N)w + J\,\frac{1}{2} Z w = w$, using $\mathbf{1}^{\mathsf{T}} w = 0$. So y is the solution we need, and

$$\det(L + J - ww^{\mathsf{T}}) = \det\big((L + J)(I - yw^{\mathsf{T}})\big) = \det(L + J)\,\big(1 - w^{\mathsf{T}} y\big).$$

In words: deleting the 2-cycle multiplies the tree count by $1 - w^{\mathsf{T}} y$, and $w^{\mathsf{T}} y$ is half the sum of the numbers $w^{\mathsf{T}} P^k w$ over $k < m$.

Each of those is a statement about returning. The strings p and q have period two, so shifting p by an even number of places can give p back and never q, and an odd shift does the opposite (Figure 2). By Lemma 3,

$$w^{\mathsf{T}} P^k w = P^k(p,p) + P^k(q,q) - P^k(p,q) - P^k(q,p) = 2\,\big(-\tfrac{1}{2}\big)^k \qquad (0 \le k < m),$$

and therefore

$$1 - w^{\mathsf{T}} y = 1 - \sum_{k=0}^{m-1} \big(-\tfrac{1}{2}\big)^k = \frac{2^{m-1} + (-1)^m}{3 \cdot 2^{m-1}}.$$

Multiplying by the de Bruijn number $2^{2^m - m - 1}$ gives case B, $\frac{1}{3}\, 2^{2^m - 2m}\big(2^{m-1} + (-1)^m\big)$, which with $m = n - 1$ is the formula in Theorem 1. Adding case A,

$$2^{2^{n-1} - 2n + 2} \left(2^{n-2} + \frac{2^{n-2} - (-1)^n}{3}\right) = \frac{2^n - (-1)^n}{3}\, 2^{2^{n-1} - 2n + 2} = T(n). \qquad \square$$

Remark. The factor $(2^{m-1} + (-1)^m)/(3 \cdot 2^{m-1})$ tends to $\frac{1}{3}$: in the limit, a third of the arborescences avoid both edges of the 2-cycle. Göttfert [6] studies the minimal polynomials of binary sequences of least period $2^N - 2$; we found no count of the cycles of length $2^n - 2$ in the literature we could reach.

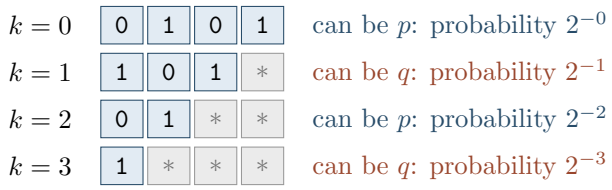


Figure 2: The walk started at $p = 0101$ ($m = 4$). After k steps the state is p shifted by k places (blue) followed by k random bits ($*$). For even k the blue part is how p starts, so the walk is at p with probability 2^{-k} and never at q; for odd k it is how q starts. By symmetry the same holds from q, which gives $w^{\mathsf{T}} P^k w = 2(-\frac{1}{2})^k$.

Verification. The counts $1, 3, 20, 2816$ for $n \leq 5$ agree with a direct enumeration of cycles, and both cases agree with exact determinants for $n \leq 11$, in two programs that share no code. Lemma 1 (in the form: the only two-edge sets whose deletion keeps $B(m)$ balanced are the two loops and the alternating 2-cycle), the effect of each deletion on the Laplacian, Lemma 3, the identity $(I - P)Z = I - J/N$, the solution y, the determinant identity, the evaluation of $w^{\mathsf{T}} P^k w$ and the closed forms are checked in Lean 4 with Mathlib [10], using only the standard axioms. The BEST theorem, the matrix-tree theorem, Lemma 2 and the de Bruijn count are cited, not formalised. Everything is at `https://github.com/jvvk/mathematics/tree/main/debruijn-cycles-two-short`.

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